

D.A.V. LOHAGHAT
CLASS TEST - II (2025-26)
SUBJECT - PHYSICS
CLASS-X

MM : 20

Time: 30 Min.

Section B - Very Short Answer Questions

- Q 1.** "The magnification produced by a spherical mirror is -3 ". List four informations you obtain from this statement about the mirror/ image. (AI 2016) [2]
- Q 2.** List two possible ways in which a concave mirror can produce a magnified image of an object placed in front of it. State the difference if any between these two images. (AI 2014) [2]

Section C - Short Answer Questions

- Q 3.** A concave mirror has a focal length of 20 cm. At what distance from the mirror should a 4 cm tall object be placed so that it forms an image at a distance of 30 cm from the mirror? Also calculate the size of the image formed. (AI 2019) [3]
- Q 4.** The image of a candle flame placed at a distance of 30 cm from a mirror is formed on a screen placed in front of the mirror at a distance of 60 cm from its pole. What is the nature of the mirror? Find its focal length. If the height of the flame is 2.4 cm, find the height of its image. State whether the image formed is erect or inverted. (Delhi 2017) [3]
- Q 5.** The image of an object formed by a mirror is real, inverted and is of magnification -1 . If the image is at a distance of 40 cm from the mirror, where is the object placed? Where would the image be if the object is moved 20 cm towards the mirror? State reason and also draw ray diagram for the new position of the object to justify your answer. (AI 2016) [3]
- Q 6.** The linear magnification produced by a spherical mirror is $+\frac{1}{3}$. Analysing this value state the (i) type of mirror and (ii) position of the object with respect to the pole of the mirror. Draw any diagram to justify your answer. (AI 2014, Foreign 2014) [3]

Section D - Long Answer Questions

- Q 7.** A student wants to project the image of a candle flame on a screen 48 cm in front of a mirror by keeping the flame at a distance of 12 cm from its pole. (AI 2014) [4]
- (a) Suggest the type of mirror he should use.
- (b) Find the linear magnification of the image produced.
- (c) How far is the image from its object?
- (d) Draw ray diagram to show the image formation in this case.